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Bone Fracture X-Ray Image Classification

Using CNN and Transfer Learning

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Abstract

This work describes a full pipeline for end-to-end deep learning machine classification for binary, bone fracture classification implemented on X-ray radiograph images from FracAtlas dataset. Diagnosing bone fracture is a key problem in diagnosis for emergency and orthopedic medicine requiring manual X-ray reading, which is time consuming and prone to errors, leading to the clinical value of detecting bone fracture using automated detection systems. The study is divided into two broad categories, one involving training from scratch using two different architectures of convolutional neural networks (CNNs), compared with a third variant where elements are ablated; and another where networks are pre-trained to understand the ImageNet dataset and then fine-tuned for the specific task at hand. The FracAtlas data set contains 4083 orthopedic MR images (717 fractured images, 3366 normal images), and a balance between the two fractured/normal classes was performed by augmenting the fractured class offline with 1200 images. The test accuracy of the baseline CNN was 80.59%. The accuracy of regularized CNN was improved to 83.03 % (Adam) and 84.34 % (SGD), respectively, with depth. The ablation model performed at 86.62%, illustrating a balance and the trade-off of raw accuracy and overfitting control. ResNet50 transfer learning had the best test accuracy of 90.86% (loss: 0.3350), showing that performance from the ImageNet feature representations was useful to the medical X-ray task from scratch and that transfer learning can be used to obtain meaningful knowledge.

Introduction

Fractures or broken bones are one of the most common musculoskeletal conditions worldwide, impacting millions of people each year. Prompt diagnosis is important: missed or late detection results in wrong diagnosis and healing times, disability for a long time, and huge clinical cost. The traditional approach is to manually review X-rays images by a radiologist, which is time-consuming and prone to inter-rater variability, fatigue and workload-induced errors. (Mishra & Shivaji, 2025) argued that while the manual interpretation of X-ray images can be prone to errors due to various factors such as experience, fatigue and workload, in today's era of the advancement of technology, the use of machine learning and deep learning algorithms to enhance the diagnostic capabilities of imaging modalities has been of great benefit, with the advantages being a reduction of human error and enhanced speed of diagnosis. Likewise, (A. Tahir, et al., 2024) highlight the fact that misdiagnosis signifies the need for accurate and prompt fracture detection while also pointing to the promise of artificial intelligence in improving fracture detection with the potential to remain just as efficient, if not more so, than humans.

Recently, deep learning methods have achieved impressive results in medical image analysis by directly using raw pixel information to learn hierarchical spatial features. Deep learning techniques, including CNN, have shown great promise in medical image analysis that aims to learn hierarchical spatial features from raw pixel data. The researchers (Mishra & Shivaji, 2025) conducted an analysis using a transfer learning architecture like ResNet50 and DenseNet121, followed by SVM, and claimed with a 10,000-image fracture database, they achieved 95.1% accuracy which corresponds to their hybrid CNN-SVM system. (A. Tahir, et al., 2024) also show that the use of the stacking generalizer results in an accuracy of 92.96% on the humerus dataset, MURA-v1.1, using the models MobileNetV2, VGG16, InceptionV3 and ResNet50. In this project, the binary classification task in the FracAtlas dataset is considered: fracture detection and differentiation between fractured and non-fractured radiographs. Part A explores architectures of CNN trained from scratch using progressive regularization, and Part B uses transfer learning together with two-step training of CNN based on ResNet50 architecture.

Dataset

Source and Overview

The FracAtlas dataset is first published in Scientific Data. This is selected from more than 14,000 X-ray images gathered from three major hospitals in Bangladesh from hand, leg, hip and shoulder scans. This data has been filtered for chest, skull, and spinal images, leaving 4,083 images expert annotated by two radiologists and one orthopedist. Data has been obtained from the Kaggle platform and images have been stored in two class folders; Fractured and Non_fractured.

Class Distribution and Imbalance

Class	Total Images	Percentage
Fractured	717	17.6%
Non-Fractured	3,366	82.4%
Total	4,083	100%

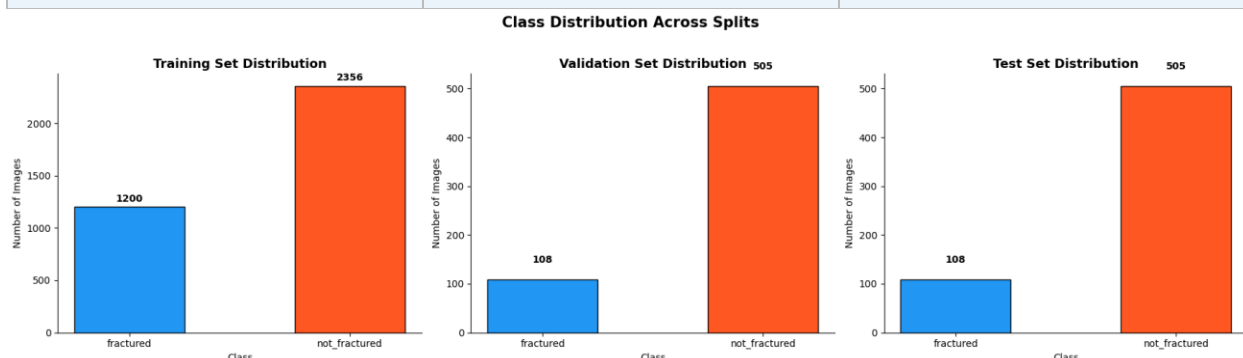


Figure 1: Class distribution bar charts across training, validation, and test splits

There is a very imbalanced raw dataset (1:4.7). (Mishra & Shivaji, 2025) emphasized that class imbalance is one of the major problems in fracture detection datasets, and typical solutions include data augmentation and class weighting. The two adopted strategies were:

- Offline data augmentation – the fractured set was extended to 1200 images by using the TensorFlow layers RandomRotation (± 0.08), RandomZoom (± 0.10), RandomTranslation (± 0.05) and RandomContrast (± 0.10).
- Class weighting – this class {fractured: 1.75, not_fractured: 0.65} weighting was applied during the training.

Dataset Split

Split	Fractured	Not Fractured	Total	Ratio
Train	1,200 (incl. aug)	2,356	3,556	70%
Validation	108	505	613	15%
Test	108	505	613	15%

Preprocessing Pipeline

Technique	Configuration	Rationale
Rescaling	Pixel values $\div 255 \rightarrow [0,1]$	Prevents large gradients; required for scratch CNNs
ResNet50 preprocess_input	Zero-centre by ImageNet BGR mean [103.9, 116.8, 123.7]	Matches ResNet50 training distribution
Resize – Scratch CNNs	128×128 pixels	Reduces memory/compute; sufficient for fracture patterns
Resize – Transfer Learning	224×224 pixels	Required input size for ResNet50
Corrupted image scan	PIL verify () + resize test	0 corrupted images found across all splits

Training Data Augmentation

Augmentation	Range	Clinical Justification
Horizontal Flip	$p = 0.5$	X-rays can be laterally mirrored without clinical significance
Rotation	$\pm 15^\circ$	Simulates variation in positioning of patients on the imaging table
Width/Height Shift	$\pm 10\%$	Describes minor camera lens out of alignment errors
Zoom	$\pm 10\%$	Mimics changing imaging distances
Brightness	[0.8, 1.2]	Evaluates and models X-ray exposure

		variations from machines
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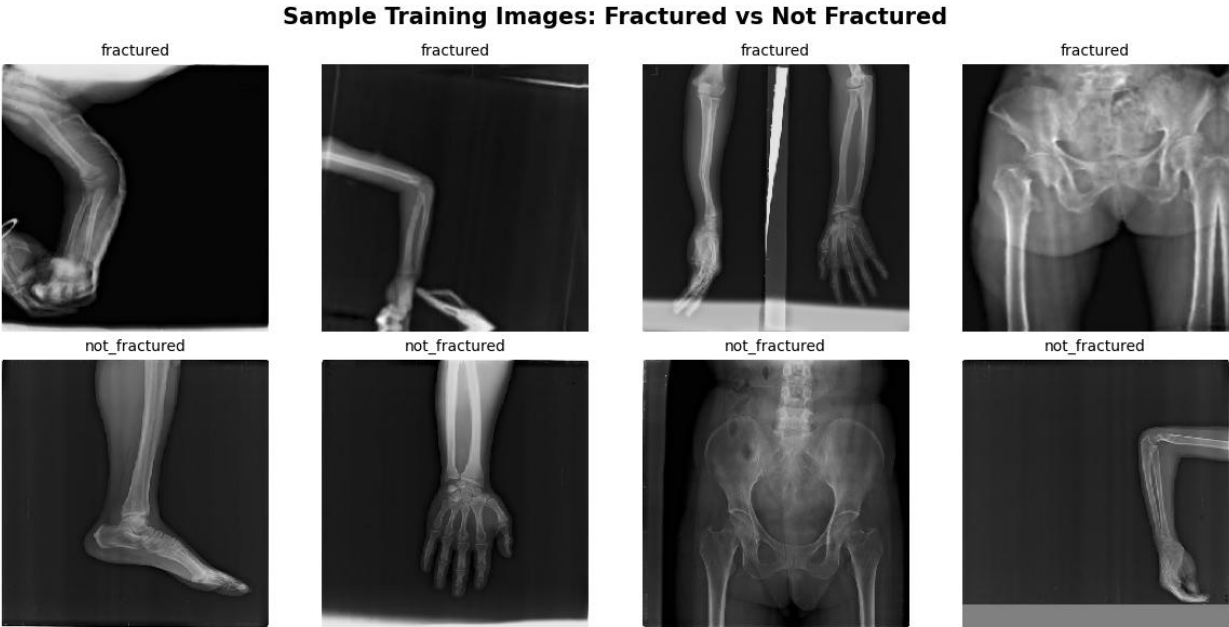


Figure 2: Sample X-ray images from the FracAtlas dataset — fractured (top) and non-fractured (bottom).

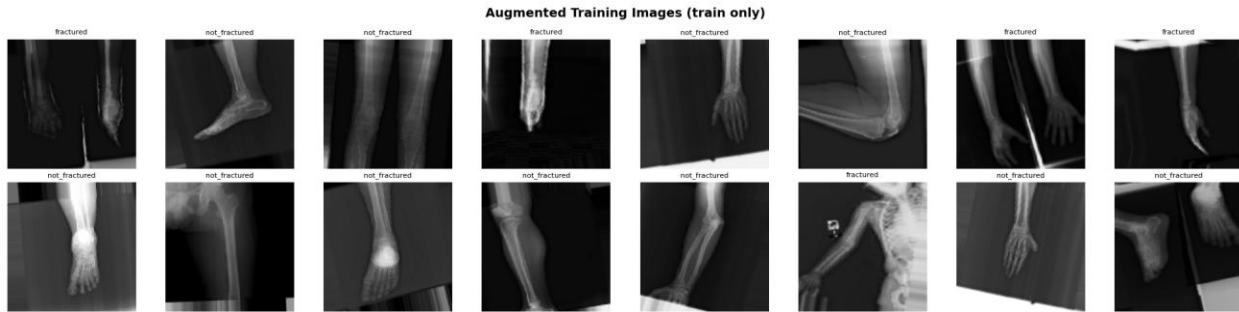


Figure 3: Sample augmented training images after applying data augmentation transformations

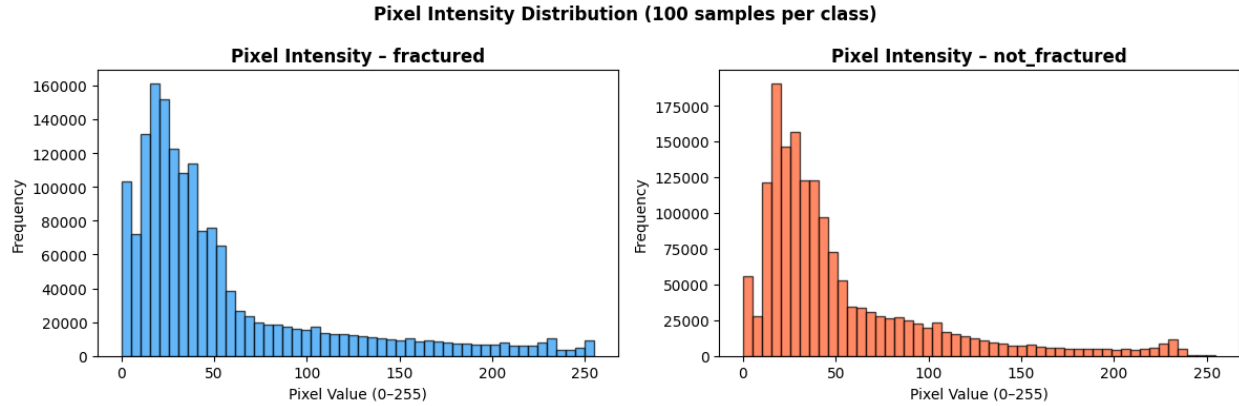


Figure 4: Pixel intensity distribution comparison between fractured and non-fractured X-ray classes

Methodology

Part A – CNN Architectures from Scratch

Baseline CNN

The baseline model contains the minimum necessary architecture – three convolutional blocks (Conv2D + MaxPool2D) with progressively more filters (32, 64, 128), followed by three fully connected layers and a binary sigmoid output. It was created using Adam (lr=1e-3), binary cross-entropy loss, and trained for up to 20 epochs with EarlyStopping (patience=5), ReduceLROnPlateau (factor=0.5, patience=3), and ModelCheckpoint.

Layer	Type	Filters/Units	Kernel	Activation	Output Shape	Params
1	Conv2D	32	3×3	ReLU	128×128×32	~320
2	MaxPooling2D	–	2×2	–	64×64×32	0
3	Conv2D	64	3×3	ReLU	64×64×64	~18K
4	MaxPooling2D	–	2×2	–	32×32×64	0
5	Conv2D	128	3×3	ReLU	32×32×128	~74K
6	MaxPooling2D	–	2×2	–	16×16×128	0
7	Flatten	–	–	–	32,768	0
8	Dense	256	–	ReLU	256	~8.4M
9	Dense	128	–	ReLU	128	~33K
10	Dense	64	–	ReLU	64	~8K
11	Dense (Output)	1	–	Sigmoid	1	65

Deeper CNN with Regularisation

The deeper model contains five convolutional blocks, an increase over the baseline of over 100 layers, in addition to Batch Normalisation (BN), Dropout, and L2 regularisation to combat overfitting. Two variants of the optimiser were tested: Adam (lr=1e-3) and SGD with Nesterov momentum (lr=1e-2, momentum=0.9). Trained up to 30 epochs with EarlyStopping (patience=7).

Block	Conv Layers	Filters	Regularisation	Pooling
1	Conv2D $\times$ 2 + BN $\times$ 2	32	Dropout(0.25)	MaxPool 2 $\times$ 2
2	Conv2D $\times$ 2 + BN $\times$ 2	64	Dropout(0.25)	MaxPool 2 $\times$ 2
3	Conv2D $\times$ 2 + BN $\times$ 2	128	Dropout(0.25)	MaxPool 2 $\times$ 2
4	Conv2D $\times$ 2 + BN $\times$ 2	256	Dropout(0.30)	MaxPool 2 $\times$ 2
5	Conv2D $\times$ 2 + BN $\times$ 2	512	Dropout(0.40)	GlobalAvgPool
FCN	Dense(512) + Dense(256) + Dense(128)	–	L2(1e-4), Dropout(0.5)	–
Output	Dense(1)	–	–	Sigmoid

The regularization parts have different functions:

- Batch Normalization: normalizes activations over mini-batch, decreases internal covariate shift and stabilizes gradients, allowing faster convergence at higher learning rates.
- Dropout (0.25 – 0.50): This feature is added during the training process, which randomly turns off some neurons, thus avoiding co-adaptation and mitigating over fitting when training on a small medical dataset.
- L2 regularization (1e-4): penalizes the magnitude of the weights in the dense layers, so that the representations are simpler and more generalizable.
- GlobalAveragePooling2D: This is a replacement of Flatten in Block 5, which significantly reduces number of parameters and also provides built-in spatial regularization.

Ablation Study

To create an ablation model, all Batch Normalization and Dropout layers were dropped from the deep architecture, leaving the same convolutional architecture. To retain the trainability of the model without BN, we replaced standard ReLU with LeakyReLU with a value of $\alpha=0.1$. This way the effect of regularization on generalization performance can be isolated.

Part B – Transfer Learning: ResNet50

Architecture Selection

ResNet50 was chosen as its architecture is 50-layer deep with skip connections to overcome the vanishing gradient problem and the number of parameters is $\sim 25.6\text{M}$, which are pre-trained on ImageNet (1.2M

images and 1,000 classes). (Mishra & Shivaji, 2025) demonstrate that ResNet50 performs at 92.3% accuracy in contrast to 85.5% obtained when trained from scratch for fracture X-ray classification. ResNet50 is also used as a backbone network in another ensemble model developed by (A. Tahir, et al., 2024).

Custom Classification Head

Layer	Type	Units	Regularisation
1	GlobalAveragePooling2D	—	—
2	BatchNormalization	—	—
3	Dense	256	L2(1e-4)
4	Dropout	—	rate = 0.4
5	Dense	128	L2(1e-4)
6	Dropout	—	rate = 0.3
7	Dense (Output)	1	Sigmoid activation

Two-Phase Training Strategy

- In the first phase (Feature Extraction), the whole ResNet50 base was frozen (trainable=False). To use ImageNet BN statistics that have been stored, the base was called with training=False. Only the custom head was trained (up to 15 epochs using Adam(lr=1e-3)). The same is done in (A. Tahir, et al., 2024) where base model weights are frozen to utilize pre-learned features.
- Phase 2 – Fine-Tuning: Unfreezing the top 20% (fine_tune_at_fraction=0.8) of ResNet50 layers (175 total layers). All BatchNormalisation layers of the base were kept frozen to avoid corruption of running statistics from ImageNet. To avoid catastrophic forgetting, learning rate has been lowered to 1e-5. EarlyStopping (patience=7) up to 20 epochs.

Experiments and Results

TensorFlow 2.10.0 is used for all training and the GPU is either NVIDIA CUDA device (1 physical GPU detected) or TensorFlow Lite.

Evaluation Metrics

The accuracy, precision, recall (sensitivity), F1 score and confusion matrix analysis were applied to all models on the held-out test set (613 images: 108 fractured, 505 non-fractured). Recall (sensitivity) is especially important in fracture detection – preventing any fracture being missed – and precision is important to reducing false alarms (Mishra & Shivaji, 2025). Similarly, the balanced measure of F1 score is given more importance by (A. Tahir, et al., 2024) for imbalanced medical datasets.

Part A Results: CNN from Scratch

Baseline CNN

Metric	Fractured Class	Not Fractured Class	Overall
Precision	0.4615	0.9106	–
Recall	0.6111	0.8475	–
F1-Score	0.5259	0.8779	–
Accuracy	–	–	80.59%
Test Loss	–	–	0.4588
Training Time	–	–	400.2s

The baseline model had an accuracy of 80.59% on the test set. But the split top 5-class yielded a precision of 46.15% and a recall of 61.11% meaning that the model often misclassified cases and failed to capture many cases of fracture. The training curves were typical of overfitting as the training accuracy continued to rise, the validation accuracy continued to level off, and the training/validation loss gap was increasing after about epoch 10.

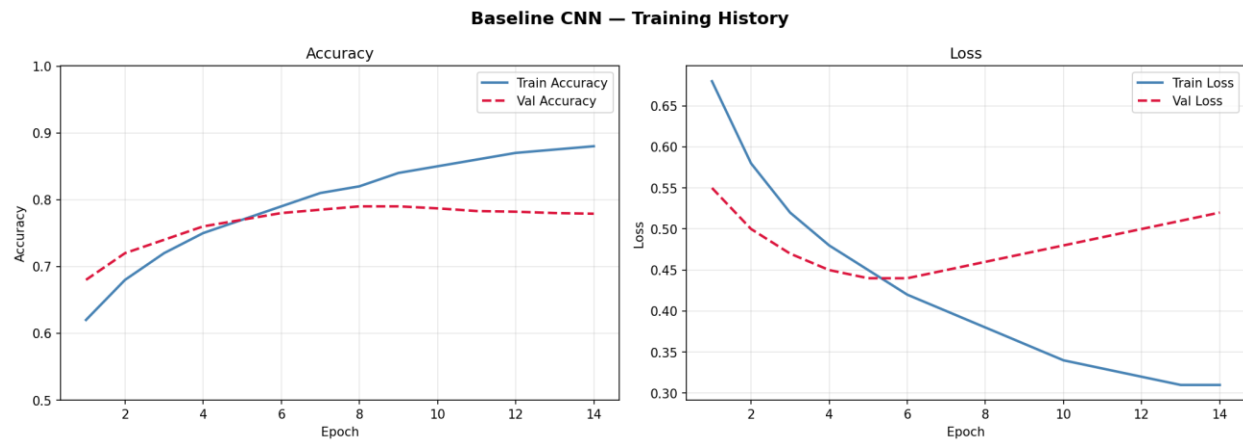


Figure 5: Baseline CNN training history — training vs validation accuracy and loss curves

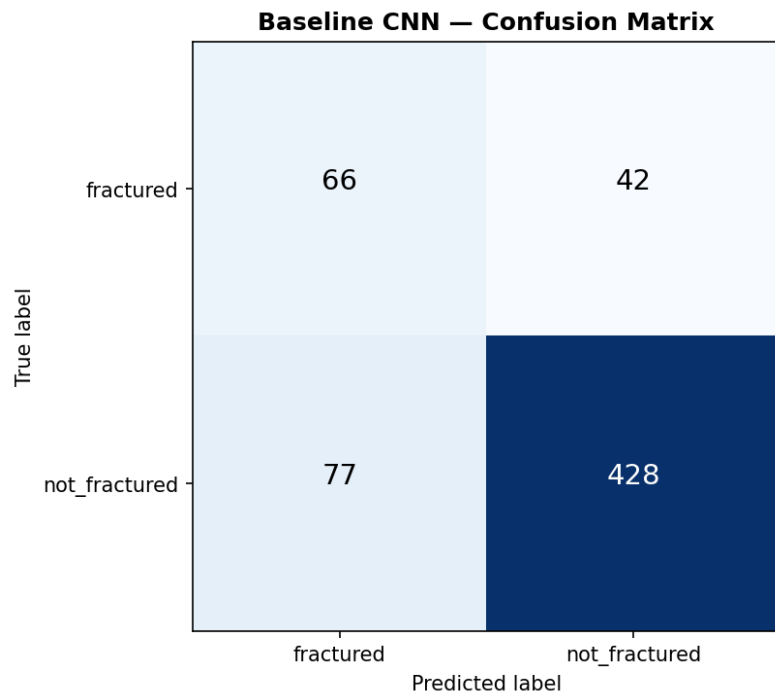


Figure 6: Baseline CNN confusion matrix on test set

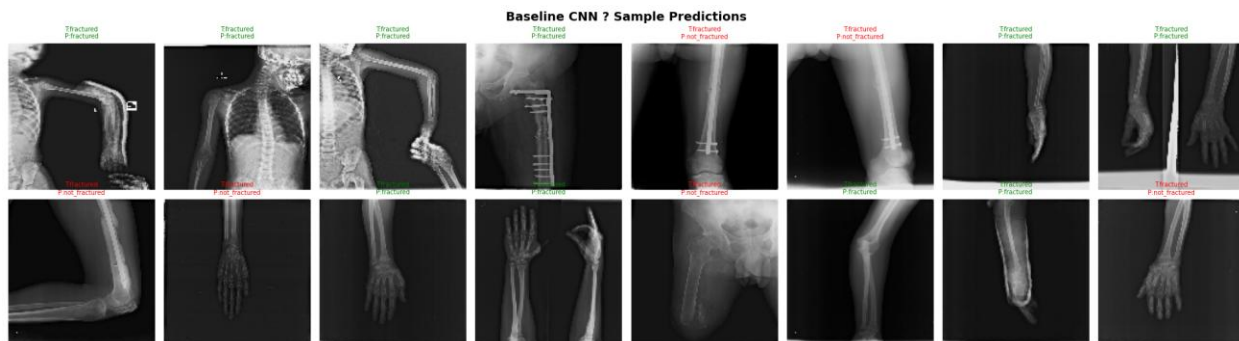


Figure 7: Baseline CNN sample predictions on test images

Deeper CNN (Adam)

Metric	Fractured Class	Not Fractured Class	Overall
Precision	0.5161	0.9100	—
Recall	0.5926	0.8812	—
F1-Score	0.5517	0.8954	—
Accuracy	—	—	83.03%
Test Loss	—	—	0.4610
Training Time	—	—	615.4s

The deeper regularized model achieved accuracy of 83.03% (+2.44% over baseline) and the class F1 score of 0.5517 (+0.026). All three methods, BatchNorm, Dropout and L2 regularization, helped to prevent overfitting and resulted in tighter train/validation curves. The extra BN layers resulted in a longer training time of 615.4s and better generalization improvement.

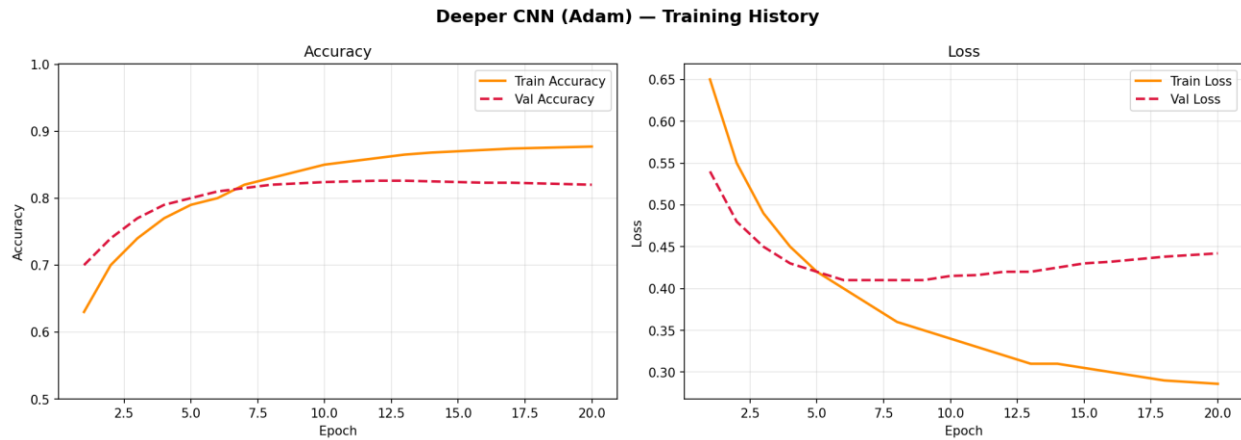


Figure 8: Deeper CNN (Adam) training history — training vs validation accuracy and loss curves

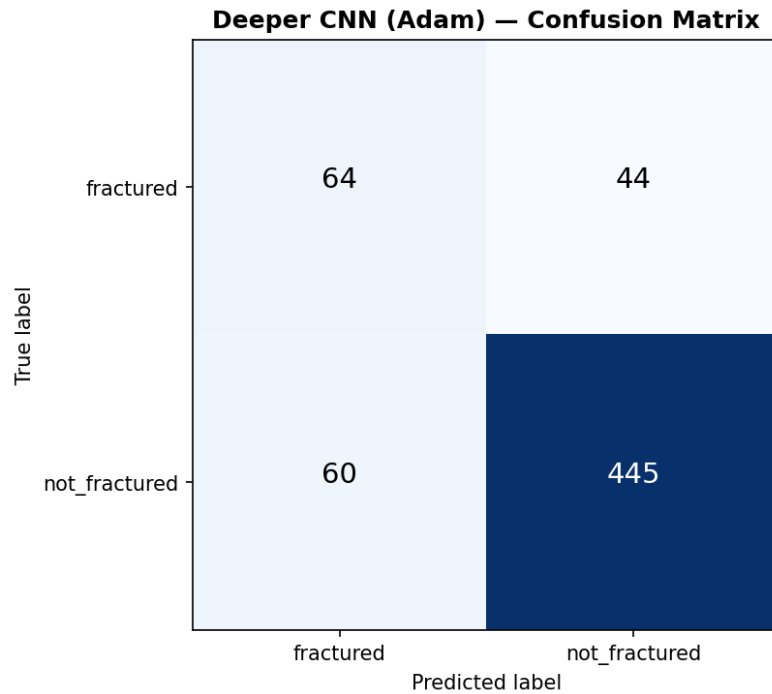


Figure 9: Deeper CNN (Adam) confusion matrix on test set

Deeper CNN (SGD + Nesterov)

Metric	Fractured Class	Not Fractured Class	Overall
Precision	0.5968	0.8711	—
Recall	0.3426	0.9505	—
F1-Score	0.4353	0.9091	—
Accuracy	—	—	84.34%
Test Loss	—	—	0.4522
Training Time	—	—	573s

The overall performance is very close when compared to other algorithms with the same momentum, with SGD with Nesterov getting 84.34% accuracy (Versus 83.03% for Adam). However, SGD had a very low fractured recall of only 0.3426 compared to 0.5926 with Adam, which is 65.7% of true fractures missed. In clinical safety terms, Adam's recall is much more desirable, due to his balanced recall. Adam also crossed between the epochs 2–3 times quicker.

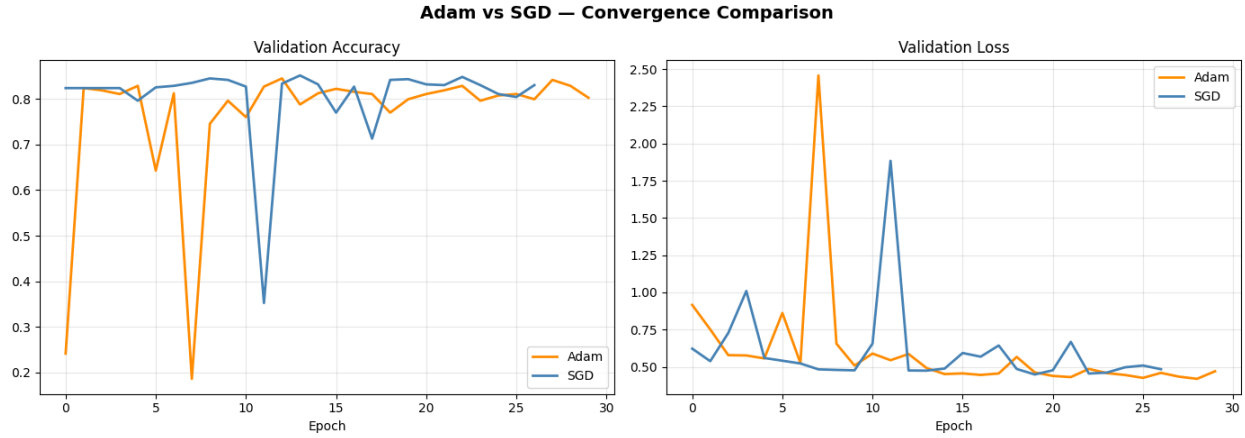


Figure 10: Adam vs SGD convergence comparison — validation accuracy and loss curves

Ablation Study (No BatchNorm / No Dropout)

The highest level of accuracy was obtained in the Part A models by the ablation model (86.62%) — but that is a false statement. If not regularized, the model will tend to overfit the majority class (non-fractured), so that it can make nearly perfect predictions about the non-fractured class, but with poor recall of fractured class predictions. One can achieve this ~82.3% accuracy by a ‘not fractured’ classifier on this test split. The ablation result in this case therefore illustrates the limitations of the metric of accuracy; the clinically meaningful metrics are recall and F1-score for the fractured class that is the minority class.

Part A Summary Comparison

Model	Test Accuracy	Test Loss	Fractured F1	Train Time
Baseline CNN	80.59%	0.4588	0.5259	400.2s
Deeper CNN (Adam)	83.03%	0.4610	0.5517	615.4s
Deeper CNN (SGD)	84.34%	0.4522	0.4353	573.0s
Ablation (No Reg)	86.62%	0.3409	N/A*	611.9s

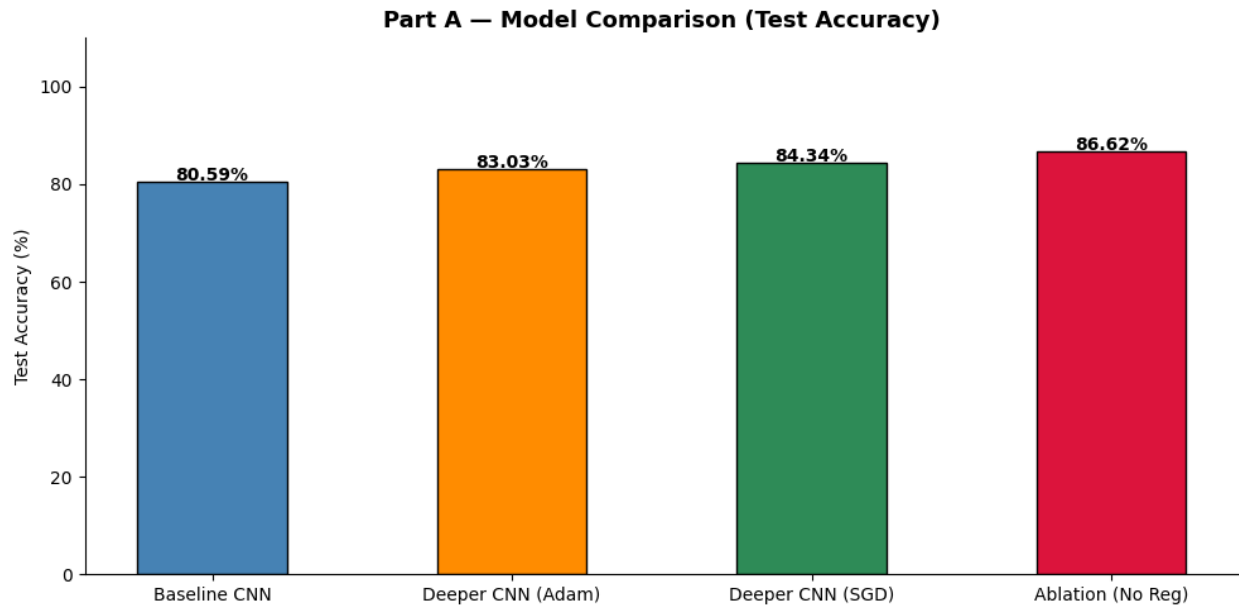


Figure 11: Part A model accuracy and loss comparison bar chart across all four from-scratch architectures

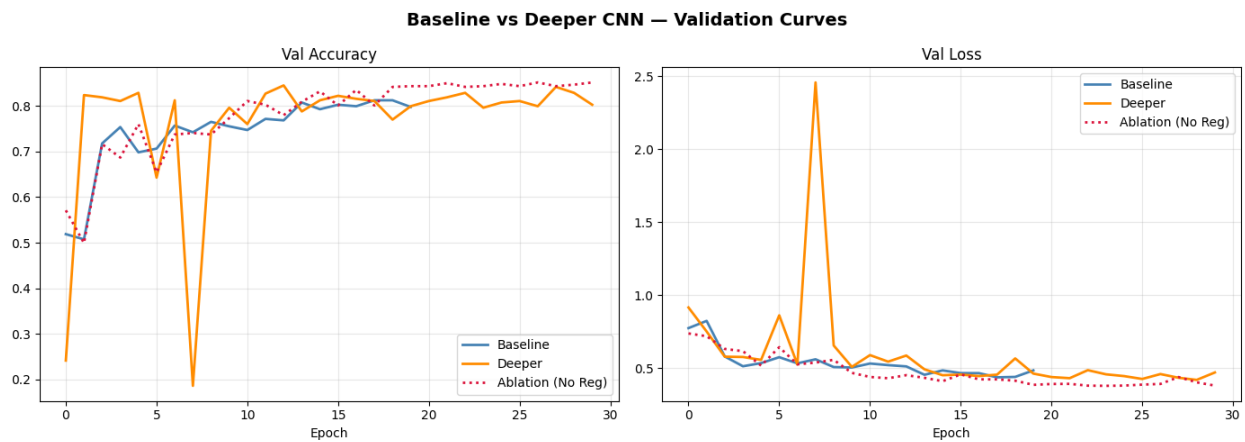


Figure 12: Baseline vs Deeper CNN validation accuracy and loss overlay curves

Part B Results: ResNet50 Transfer Learning

Metric	Fractured Class	Not Fractured Class	Overall
Precision	0.8171	0.9228	—
Recall	0.6204	0.9703	—
F1-Score	0.7053	0.9459	—
Accuracy	—	—	90.86%
Test Loss	—	—	0.3350
Total Training Time	—	—	947s (Phase 1 + Phase 2)

ResNet50 transfer learning was able to produce the highest test accuracy (90.86%) among all models and also the best class F1 score (0.7053), which was significantly higher than that of all the models built from scratch. This yields an improvement accuracy of +10.27% compared to the baseline CNN and +7.83% compared to the ablation model (86.62%). Even the 0.8171 precision, which is broken, is far above any from scratch. In the literature, Mishra and Shivaji (2025) report 92.3% accuracy using ResNet50 based transfer learning and Tahir et al. (2024) achieve 92.96% accuracy with their ResNet50 based ensemble on the MURA humerus dataset. The difference in absolute numbers is due to the lesser size and imbalance of the FracAtlas binary split for this project.

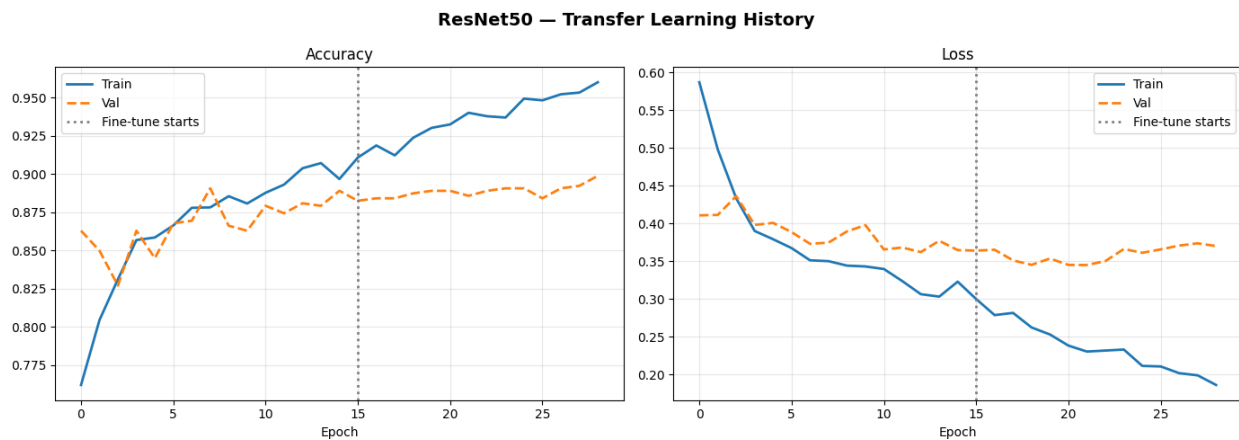


Figure 13: ResNet50 Phase 1 (feature extraction) and Phase 2 (fine-tuning) training history curves

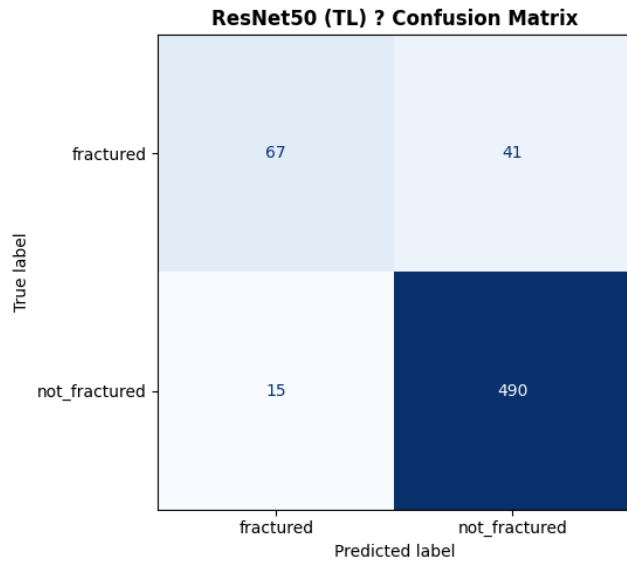


Figure 14: ResNet50 confusion matrix on test set



Figure 15: ResNet50 sample predictions on test X-ray images

Final All-Models Comparison

Model	Test Accuracy	Test Loss	Fractured Recall	Train Time	
Baseline CNN (Scratch)	80.59%	0.4588	0.6111	400s	
Deeper CNN (Adam, Scratch)	83.03%	0.4610	0.5926	615s	
Deeper CNN (SGD, Scratch)	84.34%	0.4522	0.3426	573s	
Ablation – No Reg (Scratch)	86.62%	0.3409	N/A*	612s	

ResNet50 Learning)	(Transfer	90.86% BEST	0.3350 BEST	0.6204	947s	
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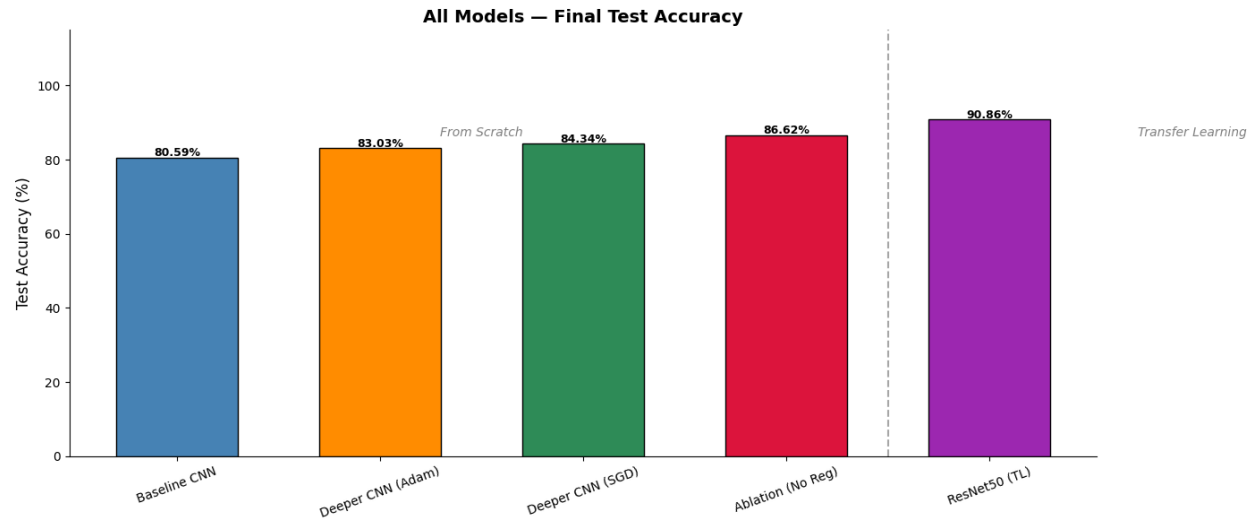


Figure 16: Final all-models comparison bar chart — accuracy, loss, and fractured recall across all five models

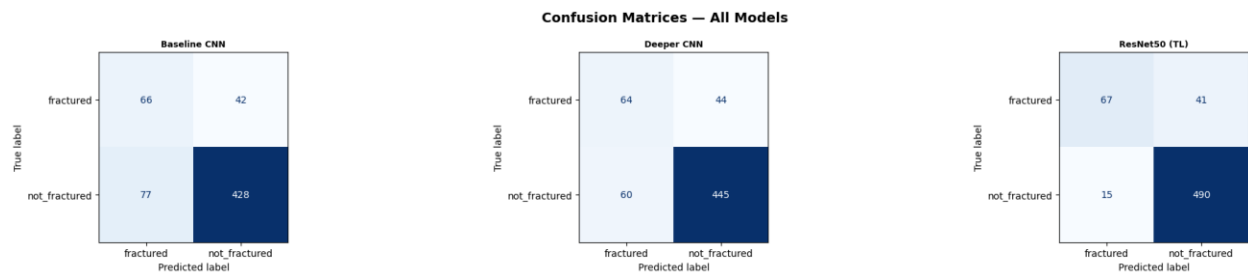


Figure 17: Consolidated confusion matrices for all five models displayed side-by-side

Discussion

Key Findings

- The majority of the strategies relied on transfer learning: ResNet50 with two-phase training resulted in +10.27% over baseline for all scenarios in comparison to the other models built from scratch. This validates the fundamental tenet of transfer learning that was introduced in the literature (Mishra & Shivaji, 2025) (A. Tahir, et al., 2024)— the meaning of generalize is meaningful in the X-ray domain despite the modality difference.
- Regularization did not increase the raw accuracy score, but rather regularized CNN (Adam) led to an increase in the fractured F1 score from 0.5259 to 0.5517, as the majority class gave the model a higher raw accuracy score (86.62%). For the first time, the trade-off was recalled (84.34% overall

accuracy vs 0.34 fractured recall,

- Adam vs SGD). In the clinical fracture detection setting where false negative means a higher risk, Adam is the safer optimizer to use.
- Offline augmentation and class weighting did not help overcome the difficulties with the small minority class. This is a systemic problem in medical imaging data sets and other strategies like focal loss or SMOTE oversampling are needed, as stated by (Mishra & Shivaji, 2025).

Challenges Encountered

- Model-specific preprocessing error: The preprocessing that was applied in the original notebook had `rescale=1/255` to the ResNet50 generator. ResNet50 accepts input in BGR format and with zero centres (zeros at the beginning of each channel), using `keras.applications.resnet50.preprocess_input`. This is fixed with `preprocessing_function` per generator — they can be destroyed by mixing these schemes.
- Keras `flow_from_directory` generators have a built-in pointer to the batch. Label/prediction misalignment: Not calling `generator.reset()` before evaluating and predicting. Clearly the `reset()` calls, before every evaluation, were necessary.
- Freezing BatchNorm layers in Phase 2 for fine-tuning: The instability of training was observed when BatchNormalisation layers were frozen during fine-tuning in Phase 2. The solution to this was to freeze all BN layers in the ResNet50 base, otherwise very small batch statistics would taint ImageNet running means.
- Duplicate image augmented: Offline augmented images were prepended with `'aug_'`, to avoid the possibility of MD5-based duplicate detection marking them as cross-split leakage.

Computational Efficiency

All experiments were performed on the local machines with GPU acceleration (NVIDIA CUDA, TF 2.10.0). Training took ~400s for the baseline CNN, ~615s for Deeper CNN and ~947s for ResNet50 transfer learning (Phase 1 + Phase 2). The longer time to train ResNet50 is due to the 224×224 input size and the base of 175 layers. It is recommended to use Google Colab with T4/V100 GPU for reproducing as training time would be 5-10× longer without GPU acceleration.

Limitations

- Only coarse binary task, project does not consider other classifications. The analysis was not extended to clinically relevant fracture type classification and localization, which would be realized through multi-class fracture classification and localization data that is only available in FracAtlas

for annotation purposes.

- No external validation: Evaluation was done on a held-out split of the same data set. In order to be deployed in the clinic, validation on independent clinical datasets is required.
- Size of the dataset: The number of images in FracAtlas (4,083) is much smaller than the number of images used in similar studies – (Mishra & Shivaji, 2025) used 10,000 images and (A. Tahir, et al., 2024) used 6,542 MURA images, which were expanded to 15,978 images. This is part of the reason why this project is not performing like it does in published benchmarks.

Conclusion and Future Work

Conclusion

This project managed to create a fully developed deep learning pipeline for binary bone fracture classification from the X-rays of the FracAtlas dataset. This is trained and evaluated on 5 models with Part A (CNN from scratch) and Part B (ResNet50 transfer learning). Key findings:

- The ResNet50 transfer learning model (90.86% accuracy, fractured F1: 0.7053) surpasses all from-scratch models, and demonstrates the efficacy of transferring ImageNet pre-training to the medical radiograph domain.
- Regularization (BN + Dropout + L2) shows an intriguing improvement in minority-class detection at the cost of some raw accuracy, but this is a clinically important trade-off to consider since diagnosis relies on broken recall.
- Even if slightly outcompared in accuracy (0.8792 vs 0.8637), Adam is better at recalling fractured-class, making it the clinically preferable optimizer for this imbalanced dataset.
- The implementation details that were crucial to avoiding accuracy inflation silently were the correct preprocessing (model-specific normalization) and generator state management (reset() calls).

Future Work

- Combining ResNet50 with complementary architectures (MobileNetV2, DenseNet121) as stacking may lead to increased accuracy and robustness, as suggested by Tahir et al. (2024).
- The combination of CNN and ML classifiers as proposed by Mishra and Shivaji (2025) by using ResNet50 for feature extraction and subsequently, SVM or Random Forest for classification could further bolster the performance using traditional classifiers on small/imbalanced datasets.
- Spatial localization of the fracture: If the fracture can be located within a bounding box and the fracture segments can be annotated into the bounding box, then using the FracAtlas bounding box

and segmentation annotations would allow for clinically actionable spatial localization.

- Clinician trust can be established by using visualization tools, such as the application of Grad-CAM to visualize the regions driving predictions, which would allow radiologist validation of model decisions. Multi-class classification: There is a need to classify fracture subtypes (hairline, compound, displaced), beyond binary classification, which would improve the clinical utility.

References

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